Modal Field: A Pre-Categorical Relational Framework for Order, Measure, Refinement, and Coarse-Grained Dynamics

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Abstract

This paper develops the *Modal Field* as a pre-categorical research programme: a formal setting in which relations, quantitative marks, additive multiplicities, transition laws, and occupancies are studied before they are identified with spacetime, matter, energy, or any other established physical category. The central question is not whether geometry can be asserted to emerge, but which structures are mathematically necessary for order, measure, refinement, probability, and macroscopic dynamics to become simultaneously coherent. The analysis combines exact enumeration, theorem-driven counterexamples, prospective computational gates, projective constructions, and stochastic controls. Exact enumeration of all $2^{20} = 1{,}048{,}576$ loopless directed graphs on five labeled vertices shows that strongly connected condensation produces a canonical partial order, but candidate scalar clocks and durations remain non-unique and uncalibrated. Global interval-abundance signatures discriminate selected Minkowski-like orders from matched nulls, yet certified local obstructions prove that these signatures are not sufficient for manifoldlikeness. A sequence of identifiability theorems then shows that order and cardinality cannot recover a preferred quantitative measure across monotone marginal transformations; an additional marked field q can break this symmetry only if it is operationally independent of the order. The global gauge $q \mapsto q + c$ preserves relative information while leaving absolute scale undetermined.

Projective refinement requires an additive base measure μ or equivalent multiplicity semantics. Exact regrouping invariance forces branch probabilities to be ratios of additive subtree weights. For the exponential family, the sufficient subtree weight is $W_\lambda(A) = \sum_{i\in A} \mu_i e^{-\lambda q_i}$, yielding an associative effective score Q_λ. This score is an exact static coarse variable for the corresponding kernel, but it is not generally an autonomous dynamical state. Gaussian closure is exact only under a Gaussian invariant family and Gaussian innovations; centering alone does not select that law. Noise-law details survive in stationary cumulants, effective scores, and operational kernels, although conditional Gaussian universality appears under independent coarse-graining with controlled exponential moments. Finally, graph-level dynamics requires an occupancy π or joint flow $F_{ij} = \pi_i K_{ij}$; q and a transition kernel alone do not determine unique macroscopic transport. The resulting minimal architecture is a proposal for disciplined pregeometry, not a completed fundamental theory. Its main contribution is a precise map of what relations can generate, what they cannot generate, and which additional structures must be independently justified.

Contents

1 Purpose and scientific position

The Modal Field is proposed as a framework for asking a narrow but foundational question:

> What mathematical information must exist before familiar physical categories such as duration, length, volume, state, transition, and flow can be defined without circularity?

The proposal belongs to the broad family of pregeometric and relational programmes in which spacetime is not assumed as a background object. Causal-set theory, dynamical graph models, tensor models, and other approaches similarly investigate the possibility that geometry is emergent rather than primitive [1–4]. The Modal Field differs in emphasis. It does not begin by postulating that a partially ordered set already represents causal spacetime, nor does it begin with a Hamiltonian designed to favor a geometric phase. It begins with a sequence of identifiability and consistency problems and admits new structure only when a theorem or counterexample shows that the previous structure is insufficient.

The term *field* is therefore not used in the ordinary sense of a function $\phi(x, t)$ defined on spacetime. No points x or times t are available at the fundamental level. A modal field is instead a distributed relational object whose state may contain:

$$\mathcal{M} = (V, R, q, \mu, K, \pi, \mathfrak{F}), \tag{1}$$

where V is a finite or countable set of alternatives, R is a directed relation, q is a quantitative mark, μ is an additive base measure, K is a transition kernel, π is an occupancy, and $\mathfrak{F}$ denotes a refinement structure. Equation (1) is an architectural summary, not a declaration that all seven objects are fundamental. A central objective is to determine which of them can be derived from the others and which remain independent inputs.

The programme is meant to be generative rather than merely negative. It supplies constructions that are exact under declared assumptions, including gauge-invariant transition laws, projective measures on finite and infinite refinement trees, associative partition-sum messages, and exact flow aggregation. At the same time, it treats failed gates and no-go results as primary results. This balance is essential because a pregeometric framework that can absorb every obstruction by adding an unconstrained primitive would cease to make determinate claims.

1.1 What this research is for

The immediate use of the research is methodological and structural. It provides:

(i) a catalogue of minimal data needed to pass from relations to quantitative observables;

(ii) exact criteria for projective consistency under changes of resolution;

(iii) operational tests separating gauge-invariant relative information from uncalibrated absolute quantities;

(iv) diagnostics for whether a coarse variable is sufficient only statically or also dynamically;

(v) explicit failure modes for unjustified claims of emergent spacetime.

A longer-term physical use would require a bridge from the modal variables to experimentally accessible preparations and measurements. That bridge is not supplied here, but the formal constraints established below narrow what such a bridge would have to accomplish.

1.2 Common misreadings

Three misreadings are particularly damaging. First, an internal partial order is not automatically physical time. Second, a symmetric graph cost is not automatically spatial distance. Third, a log-partition expression is not automatically thermodynamic free energy. Mathematical resemblance can motivate a hypothesis, but it cannot replace calibration, operational definition, or empirical identification. The paper therefore uses terms such as *internal order*, *computational relational cost*, and *effective score* until stronger identifications are earned.

2 Relation to established work

The causal reconstruction results of Hawking, King, McCarthy, and Malament show that, under appropriate continuum assumptions, causal structure determines the conformal geometry of a distinguishing spacetime [5, 6]. Causal-set theory adds local finiteness, motivating the slogan that order plus number can recover geometry [1, 2]. These results are a major point of reference, but their assumptions matter. They concern structures already known to belong to a suitable Lorentzian class or to approximate one. They do not imply that an arbitrary directed relation generates a Lorentzian manifold.

The Modal Field programme investigates the inverse problem from a deliberately weaker starting point. Its early audits ask what can be extracted from generic finite relations; later audits ask what extra quantitative structure is necessary when order alone is non-identifying. This makes the programme complementary to manifold-reconstruction work, interval-abundance diagnostics [7], dimension estimators [8], and embedding algorithms [9]. The emphasis on explicit counterexamples is also consistent with the wider causal-set concern that generic finite orders are usually not manifoldlike [2, 10].

Projective consistency connects the framework to probability theory and stochastic processes. Consistent finite-dimensional marginals can define measures on infinite path spaces through extension theorems [11–13]. The use of relative entropy and exponential families follows classical information-theoretic variational methods [14–17]. The coarse-graining of transition kernels is related to Markov aggregation and lumpability [18–21]. These connections are used as mathematical tools; they do not imply that the Modal Field is already a statistical-mechanical or Markovian physical theory.

3 Formal setup and notation

Definition 1 (Relational substrate)**.** *Let $V = \{1, \dots, n\}$ and let $R \subseteq V \times V$ be a loopless directed relation. Its adjacency matrix is $A \in \{0, 1\}^{n\times n}$ with $A_{ii} = 0$.*

The relation need not initially be transitive or antisymmetric. Let $\mathrm{SCC}(R)$ denote the partition into strongly connected components. Contracting each component gives the condensation graph $C(R)$, which is a directed acyclic graph. Its transitive closure induces a partial order:

$$[X] \preceq_R [Y] \quad \Longleftrightarrow \quad [X] = [Y] \text{ or there is a directed path from } [X] \text{ to } [Y]. \tag{2}$$

Here $[X]$ and $[Y]$ denote strongly connected components. Equation (2) defines an internal precedence relation without interpreting it as physical chronology.

Definition 2 (Quantitative mark and gauge)**.** *A mark q is a real-valued assignment to nodes, edges, or alternatives, according to the audited construction. The primary gauge transformation is*

$$q_\alpha \longmapsto q_\alpha + c, \qquad c \in \mathbb{R}, \tag{3}$$

where the same constant is added to all marks in the relevant connected context.

For edge marks, define a positive cost

$$\ell_{ij} = e^{q_{ij}}. \tag{4}$$

For vertices connected by directed paths, let $d_{\text{dir}}(i, j)$ be the minimum sum of ℓ along a directed path from i to j. A symmetric round-trip cost is

$$D_{\text{eff}}(i, j) = \frac{1}{2}\left[d_{\text{dir}}(i, j) + d_{\text{dir}}(j, i)\right]. \tag{5}$$

Under (3), every path cost scales by e^c, hence

$$D_{\text{eff}} \longmapsto e^c D_{\text{eff}}. \tag{6}$$

Any normalization by a positive homogeneous scale functional $S(D)$,

$$\widehat{D} = \frac{D_{\text{eff}}}{S(D_{\text{eff}})}, \qquad S(aD) = aS(D), \tag{7}$$

is gauge invariant. The audited conclusion is therefore an effective computational geometry up to global scale, not a calibrated physical metric.

Definition 3 (Base measure). *A base measure is a positive assignment $\mu_i > 0$ or $\mu_{ij} > 0$ satisfying additivity under declared refinement. If a macro-alternative A is partitioned into children B_a, then*

$$\mu(A) = \sum_a \mu(B_a). \tag{8}$$

Definition 4 (Transition and occupancy). *A row-stochastic kernel K satisfies*

$$K_{ij} \geq 0, \qquad \sum_j K_{ij} = 1. \tag{9}$$

An occupancy π is a probability distribution on source states,

$$\pi_i \geq 0, \qquad \sum_i \pi_i = 1. \tag{10}$$

The joint flow is

$$F_{ij} = \pi_i K_{ij}. \tag{11}$$

The distinction between K and π becomes essential in graph coarse-graining.

4 Computational epistemology

The computational programme was designed around prospective gates, exact controls, and reproducible seeds. A result was classified as one of four types:

(a) an exact theorem or algebraic identity;

(b) an exhaustive finite enumeration;

(c) a prospective computational test with frozen thresholds;

(d) a counterexample proving non-sufficiency.

The distinction matters. A simulation can verify implementation and characterize a declared ensemble, but it cannot establish that nature uses that ensemble. Conversely, one exact counterexample is sufficient to disprove a universal closure claim even if the counterexample is rare.

Three corrective audits are reported rather than hidden. A36 originally used a finite-size Kolmogorov–Smirnov threshold that was too strong for lattice-valued Rademacher sums; A36.1 replaced the diagnostic with convergence of characteristic functions while preserving the central-limit assumptions. A37 accumulated round-off error in an infinite Gaussian log-MGF series; A37.1 used the exact Gaussian closed form. A38 used a prevalence threshold to test a universal median-closure property; A38.1 replaced it with a decisive exact counterexample. These corrections changed diagnostics, not the underlying scientific conclusions.

Table 1: Symbols used throughout the paper. A symbol is not assigned a physical interpretation beyond the statement in the right column.

Symbol	Meaning
V	Finite or countable set of modal alternatives.
R, A	Directed relation and adjacency matrix.
$C(R)$	Condensation DAG obtained by contracting strongly connected components.
$\preceq_R$	Partial order induced by reachability in $C(R)$.
q	Additional quantitative mark; not assumed to be distance, energy, or time.
μ	Additive base measure or multiplicity; not assumed to be physical volume.
λ, β	Dimensionless coupling strengths in audited exponential kernels.
K	Row-stochastic transition kernel.
π	Occupancy of source states.
$F = \mathrm{diag}(\pi)K$	Joint transport flow.
Q_λ	Log-partition effective score; not identified with free energy.
$\mathfrak{F}$	Refinement hierarchy or projective family.

5 From generic directed relations to internal order

5.1 Residual structure beyond degree sequences

The early exact audits began with all loopless directed graphs on five labeled vertices. There are

$$2^{5(5-1)} = 2^{20} = 1{,}048{,}576 \tag{12}$$

such graphs. Grouping them by in-degree and out-degree data produced 225,025 labeled degree fibers. Of the 136,793 fibers containing more than one graph, 93,167 retained residual variation under the first audited structural profile, and 95,053 varied under the full profile. Degree data therefore do not determine the relational organization.

Finite projectivity also failed to select a unique graph law. For unlabeled loopless digraph classes up to $N = 4$, the induced-subgraph kernels composed exactly, but the family of projective distributions up to a fixed top size N remained affinely isomorphic to the full simplex on N-vertex isomorphism classes. Projectivity constrained marginal consistency without selecting a measure. Reversal and complement symmetries reduced the simplex dimension but still left broad families. This is the first appearance of a recurring pattern:

$$\text{consistency} \;\not\Longrightarrow\; \text{uniqueness}. \tag{13}$$

5.2 Canonical condensation order

The A8 audit enumerated the same 1,048,576 graphs and found 5,234 distinct labeled condensation-poset codes. Of the graphs, 565,080 were strongly connected and 483,496 had nontrivial condensations. Reversing every edge preserved the SCC partition, height, width, and gradedness while exchanging sources and sinks. A single edge flip changed the full labeled condensation order in 29.296875% of cases. The condensation construction therefore yields a canonical and nontrivial internal order, while also displaying sensitivity to local relational changes.

Proposition 1 (Internal order)**.** *For every finite directed relation R, the reachability relation on $C(R)$ defined by* (2) *is a partial order.*

This proposition is elementary but foundational. It establishes an order without assuming acyclicity at the microscopic level. It does not establish a preferred temporal orientation because edge reversal produces the dual order while preserving many intrinsic invariants.

5.3 Clock candidates and their non-uniqueness

Three scalar clocks were tested on the 5,234 unique condensation posets:

$$T_{\text{depth}}(x) = \frac{\text{longest-chain depth of } x}{\text{poset height}}, \tag{14}$$

$$T_{\text{bal}}(x) = \frac{d_{\downarrow}(x)}{d_{\downarrow}(x) + d_{\uparrow}(x)}, \tag{15}$$

$$T_{\text{LE}}(x) = \frac{\mathbb{E}[\text{pos}_L(x)] - 1}{|P| - 1}, \tag{16}$$

where $d_{\downarrow}$ and $d_{\uparrow}$ are longest downward and upward depths, and the expectation in (16) is over uniform linear extensions L of the poset. All three clocks were strictly increasing on comparable pairs. However, their injectivity on nontrivial posets was only 4.97%, 34.09%, and 47.85%, respectively, and 10.38% of nontrivial posets assigned identical normalized values under all three clocks. On incomparable pairs, the depth and balance clocks disagreed in 41.76% of cases.

The result is not that internal clocks are impossible. The result is that order alone does not select a unique scalar clock. A physical clock would require an additional calibration or dynamical principle.

5.4 Ordinal durations

For comparable $x \prec y$, three interval durations were examined: the longest-chain length $L(x, y)$, the interval cardinality $V(x, y)$, and a cover-potential difference when a rank potential exists. Exact additivity along comparable triples $x \prec y \prec z$ requires

$$\tau(x, z) = \tau(x, y) + \tau(y, z). \tag{17}$$

Across all comparable triples, L satisfied (17) in 98.9282% of cases, while interval volume did so in 81.4577%. The longest-chain candidate is therefore the strongest audited ordinal duration, but it is not a calibrated proper time. A coarse-graining counterexample showed that no single multiplicative factor preserves all longest-chain durations under quotienting.

6 Minkowski-like signatures and the failure of sufficiency

The next stage tested whether interval statistics can identify orders compatible with sprinklings into finite Alexandrov intervals of 1 + 1-dimensional Minkowski spacetime. This direction is motivated by causal-set manifoldlikeness and interval-abundance work [7–9].

Let $N_m(P)$ count order intervals containing exactly m internal elements, and define the normalized profile

$$\widehat{n}_m(P) = \frac{N_m(P)}{\sum_{r=0}^{m_{\max}} N_r(P)}. \tag{18}$$

A covariance-aware statistic compared the observed vector $\widehat{\boldsymbol{n}}$ with a Minkowski reference mean $\boldsymbol{\mu}_n$ and covariance Σ_n:

$$Q(P) = (\widehat{\boldsymbol{n}}(P) - \boldsymbol{\mu}_n)^T \, \Sigma_n^{-1} \, (\widehat{\boldsymbol{n}}(P) - \boldsymbol{\mu}_n) \, . \tag{19}$$

The A14 prospective test passed all nine gates, rejecting selected ordering-fraction-matched null families while retaining Minkowski holdouts.

The positive result was then deliberately attacked. A15 inserted certified induced S_3 order-dimension obstructions into otherwise compatible structures. Many candidates remained accepted by the global A14 statistic while being provably incompatible with exact two-dimensional product-order representation. Figure 1 shows the resulting separation between a useful global signature and a sufficient manifoldlikeness criterion.

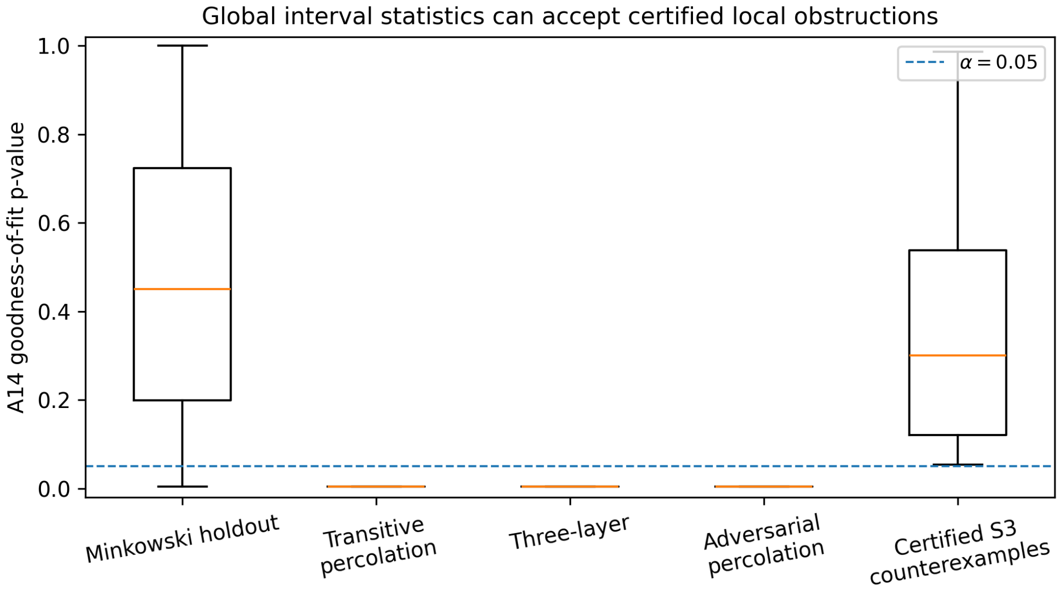

Figure 1: Real audit data from A14 and A15. The covariance-aware interval statistic rejects the matched null families but can assign large p-values to structures containing a certified induced S_3 obstruction. The result establishes that a global interval signature can be useful and still be non-sufficient.

A16 combined global signatures with exact local obstruction tests but failed one of ten prospective gates because clustered exact product orders remained difficult to classify. A17 added multiscale interval self-similarity and rejected some heterogeneous constructions, yet compact density clusters often passed. These results motivate a strict distinction:

$$\text{Minkowski-like statistics} \;\not\Longleftarrow\; \text{exact manifold embedding.} \tag{20}$$

The Modal Field therefore retains interval statistics as necessary-model diagnostics rather than promoting them to a derivation of spacetime.

7 Order-only identifiability limits

7.1 Monotone marginal symmetry

Consider latent pairs (X_i, Y_i) and the product order

$$i \prec j \quad \Longleftrightarrow \quad X_i < X_j \text{ and } Y_i < Y_j. \tag{21}$$

For strictly increasing functions f and g,

$$X_i < X_j \iff f(X_i) < f(X_j), \qquad Y_i < Y_j \iff g(Y_i) < g(Y_j). \tag{22}$$

Hence the entire order matrix is invariant under independent monotone marginal transformations. A18 tested 270 paired transformations and obtained exact order identity in every pair. The computational result is a finite witness of a general theorem.

Theorem 1 (Marginal non-identifiability)**.** *For iid continuous samples with fixed copula C, every choice of strictly increasing continuous marginals induces the same probability law on finite product orders. Consequently, no estimator using only order matrices and cardinalities can identify the marginal distributions.*

The theorem follows from the probability integral transform and Sklar's representation [22, 23]. A19 extended the argument from a single order to the full iid ensemble: all cylinder probabilities of the order-valued sequence remain identical. This is not a claim that order contains no geometry. It is a precise statement that order does not identify the marginal or conformal-volume representative without additional quantitative information.

7.2 Endogenous dynamics does not break the symmetry

Suppose the full state and update law are measurable functions of the relational history and model-independent randomness:

$$S_{t+1} = \Phi_t(S_{0:t}, U_t), \qquad S_0 = \Psi(R_0, U_{-1}). \tag{23}$$

If two latent representatives produce the same initial relation, an equivariant coupling using the same U_t produces identical relational trajectories. A21 verified this for deterministic and stochastic updates and formulated the following no-go:

$$\text{order-measurable dynamics cannot select a non-order-measurable calibration.} \tag{24}$$

Birth order, growth filtration, rank marks, and independent marks therefore do not solve the identifiability problem. A binary calibrated mark is already mathematically sufficient to break the degeneracy, but its threshold must be independently fixed. The resource is calibration, not alphabet size.

8 The role and status of the quantitative field q

8.1 Information criterion

The RZS quantitative field can escape the order-only no-go only if it is not itself reconstructible from the order and model-independent randomness. Formally, for latent model label Z,

$$\mathcal{L}(q \mid R, Z = 0) \neq \mathcal{L}(q \mid R, Z = 1) \tag{25}$$

must hold for at least one order event of positive probability. If instead $q = F(R, U)$ with the same conditional law in all latent representatives, it adds no symmetry-breaking information.

8.2 Gauge-equivariant dynamics

The audited centered update is

$$q_{t+1} = q_t - \frac{1}{2}\,\mathrm{center}(\eta\ \mathrm{center}(q_t) + \xi_t)\,, \tag{26}$$

where $\mathrm{center}(x) = x - \bar{x}\mathbf{1}$. Under $q_t \mapsto q_t + c\mathbf{1}$, the centered term is unchanged, so

$$q_{t+1} \mapsto q_{t+1} + c\mathbf{1}. \tag{27}$$

The constant mode is neither damped nor selected. In the noiseless centered sector, the contraction factor is

$$a = 1 - \frac{\eta}{2}. \tag{28}$$

Writing the same contraction as a continuous relaxation gives

$$a = e^{-\gamma\Delta\tau}, \qquad \gamma\Delta\tau = -\log a. \tag{29}$$

Only the product of rate and step duration is identified. Without an independently calibrated $\Delta\tau$, η is a per-step parameter, not a physical decay rate.

8.3 Operational transition witness

A local gauge-invariant transition law is

$$K_{ij}(q) = \frac{A_{ij}e^{-\beta q_{ij}}}{\sum_k A_{ik}e^{-\beta q_{ik}}}. \tag{30}$$

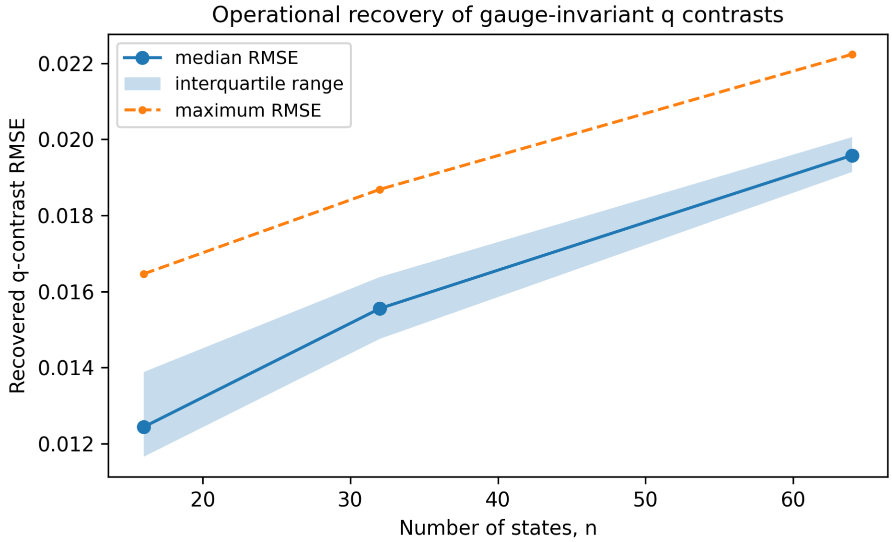

Figure 2: A24.1 contrast-recovery audit. Log-odds generated from the local exponential transition law recover gauge-invariant q contrasts with low RMSE. This is an operational witness conditional on the kernel and calibrated β, not a derivation of the physical coupling.

For two available destinations j and k,

$$\log \frac{K_{ij}}{K_{ik}} = -\beta \left(q_{ij} - q_{ik} \right) . \tag{31}$$

Equation (31) shows that transition frequencies can operationalize q contrasts if β is calibrated. The A24.1 audit recovered contrasts with mean RMSE 0.01599 and maximum RMSE 0.02224, while gauge, locality, and relabeling errors remained at floating-point precision. Figure 2 reports the recovery as a function of system size.

The same equation also exposes a scale degeneracy. Replacing q by aq and β by β/a leaves K unchanged. Raw q magnitude and coupling strength are therefore not separately identifiable without a scale convention.

9 Kernel selection, score invariance, and the free coupling

9.1 Scale-free local shapes

A local standardized score can be defined by

$$z_{ij} = \frac{q_{ij} - \overline{q}_i}{s_i}, \qquad s_i^2 = \frac{1}{d_i} \sum_{k:A_{ik}=1} (q_{ik} - \overline{q}_i)^2. \tag{32}$$

Positive affine transformations $q \mapsto aq + b$ leave z unchanged for $a > 0$. Any positive function f yields a normalized local kernel

$$K_{ij}^{(f)} = \frac{A_{ij} f(z_{ij})}{\sum_k A_{ik} f(z_{ik})}. \tag{33}$$

A25 proved that infinitely many choices of f satisfy locality, relabeling equivariance, and scale freedom while producing different transition laws. Structural admissibility therefore does not select a unique coupling.

9.2 Exponential-family selection

A stronger axiom can select the functional family. Suppose odds depend only on score differences:

$$\frac{K_j}{K_k} = G(z_j - z_k), \tag{34}$$

and independent score differences compose additively. Then

$$G(x+y) = G(x)G(y). \tag{35}$$

Continuity and positivity imply

$$G(x) = e^{-\lambda x}, \tag{36}$$

so

$$K_j = \frac{e^{-\lambda z_j}}{\sum_k e^{-\lambda z_k}}. \tag{37}$$

A26 showed that independence of irrelevant alternatives, detailed balance, and aggregation alone do not force (37); the difference-only odds functional equation does. Maximum relative entropy reaches the same family after an expected-score constraint is supplied [14, 15, 17].

The parameter λ remains free. A27 found that it is absorbable into an unstandardized raw score, but becomes an identifiable dimensionless model parameter after a fixed standardization convention. Synthetic transition data estimated $\lambda = 0.5, 1, 2$ with RMSEs 0.00401, 0.00444, and 0.00861, respectively. Identifiability from accepted data is not derivation from first principles.

9.3 Context–scale incompatibility

A28 established a no-go for a context-independent, scale-free, continuous cardinal score. Let $D_S(x, y)$ denote the score difference between alternatives x and y in context S. Assume:

$$D_{S\cup T}(x,y) = D_S(x,y) \qquad \text{for arbitrary extensions } T, \tag{38}$$

$$D(ax+b, ay+b) = D(x,y) \qquad \text{for all } a > 0, \tag{39}$$

$$D(x,x) = 0, \quad D \text{ continuous}. \tag{40}$$

Extension stability removes context dependence. Translation invariance yields $D(x, y) = h(x - y)$, and scale invariance gives $h(ad) = h(d)$. Continuity at zero forces

$$D(x,y) \equiv 0. \tag{41}$$

A nontrivial score must therefore give up at least one of extension stability, full affine invariance, continuity/cardinality, or independent scale freedom. This result explains why local z-scores, ranks, raw differences, and global normalizations exhibit different trade-offs rather than one candidate dominating all others.

10 Refinement semantics and additive measure

10.1 Why clone refinements require mass

Suppose an alternative r is replaced by exact clones $r_1, \ldots, r_m$ with the same mark q_r. If every listed item is assigned unit weight, its aggregate probability is multiplied by m. This confuses descriptive refinement with ontic multiplication.

Represent a marked context as

$$\nu = \sum_j \mu_j \delta_{q_j}. \tag{42}$$

For a positive response function f and context score T_ν, define

$$P_j = \frac{\mu_j f(T_\nu(q_j))}{\sum_k \mu_k f(T_\nu(q_k))}. \tag{43}$$

If the clone masses satisfy

$$\sum_{a=1}^{m} \mu_{r_a} = \mu_r, \tag{44}$$

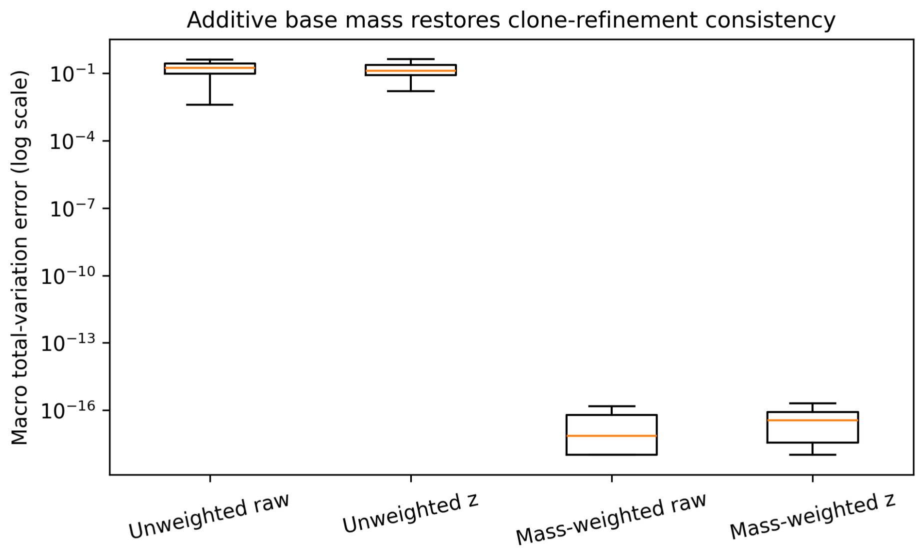

Figure 3: A29.1 clone-refinement audit. Unit counting changes macro probabilities when an alternative is descriptively cloned. Additive masses preserve the marked measure and restore exact projectivity, both for raw and context-normalized scores. Values at the numerical floor correspond to exact results up to floating-point precision.

then ν is unchanged and the aggregate clone probability equals the original probability exactly. A29.1 verified this theorem across 24,000 refinement cases. Figure 3 shows the sharp difference between unweighted and mass-weighted rules.

Weighted z-normalization remains singular when the weighted variance vanishes. Continuity under adding a vanishing-mass alternative holds for fixed positive variance but is not uniform near the zero-variance boundary. The theorem therefore requires an explicit degeneracy convention.

10.2 Semantic no-go

A30 proved that a bare refined graph with identical q marks cannot distinguish two interpretations:

(i) m descriptive copies of one macro-alternative, whose masses should sum to the original mass;

(ii) m ontically distinct states, each carrying a separate counting unit.

Because the intrinsic relational data are identical, no deterministic rule of (R, q) alone can choose between the required mass assignments. The conclusion is not that additive measure is invalid. It is that the semantics of multiplicity are extra information.

10.3 Finite refinement trees

Let terminal leaves l have positive weights w_l. The unique additive extension is

$$\mu(A) = \sum_{l \in \mathrm{Desc}(A)} w_l. \tag{45}$$

Every additive state of the form

$$S_f(A) = \sum_{l \in \mathrm{Desc}(A)} w_l f(q_l) \tag{46}$$

is independent of the order of merging. A31 audited 2,500 random trees with six merge orders each; the maximum additive-state error was 4.55×10^{-13}. Independent conservative clone splits commuted exactly within numerical tolerance. Equal leaf weights reduce (45) to counting measure, conditional on the terminal-leaf ontology.

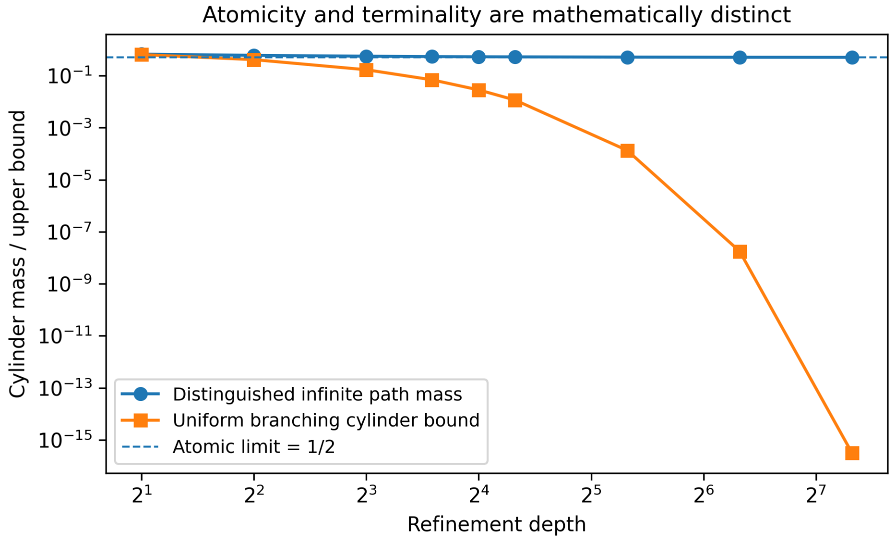

Figure 4: A32 exact witnesses. The telescoping distinguished path converges to mass $1/2$ without terminating, while a uniform contraction bound forces all cylinder masses to zero. Terminality and atomicity are therefore independent properties.

10.4 Infinite refinement and nonterminal atoms

Terminal leaves are not mathematically necessary. At each node, assign positive child fractions $p_{A,a}$ satisfying

$$\sum_a p_{A,a} = 1. \tag{47}$$

The mass of a depth-d cylinder $[a_1 \cdots a_d]$ is

$$\mu([a_1 \cdots a_d]) = \prod_{r=1}^{d} p_{a_1 \cdots a_{r-1}, a_r}. \tag{48}$$

Consistency across levels defines a probability measure on infinite path space by the standard extension theorem [11–13].

Atomicity is distinct from terminality. Along a distinguished path, choose

$$p_n = 1 - \frac{1}{n^2}, \qquad n \geq 2. \tag{49}$$

Then

$$\prod_{n=2}^{N} \left(1 - \frac{1}{n^2}\right) = \frac{N+1}{2N} \longrightarrow \frac{1}{2}. \tag{50}$$

An infinite nonterminal path therefore carries atom mass $1/2$. Conversely, if all child fractions are bounded by $r < 1$, every depth-d cylinder is at most r^d and the measure is non-atomic. Figure 4 shows both witnesses.

A scalar mark process on the refinement filtration can converge without terminality. If

$$Q_{d+1} = Q_d + \epsilon_d, \qquad \mathbb{E}[\epsilon_d \mid \mathcal{F}_d] = 0, \tag{51}$$

and independent increments satisfy

$$\sum_d \mathrm{Var}(\epsilon_d) < \infty, \tag{52}$$

then (Q_d) is Cauchy in L^2. Constant-size increments fail this criterion.

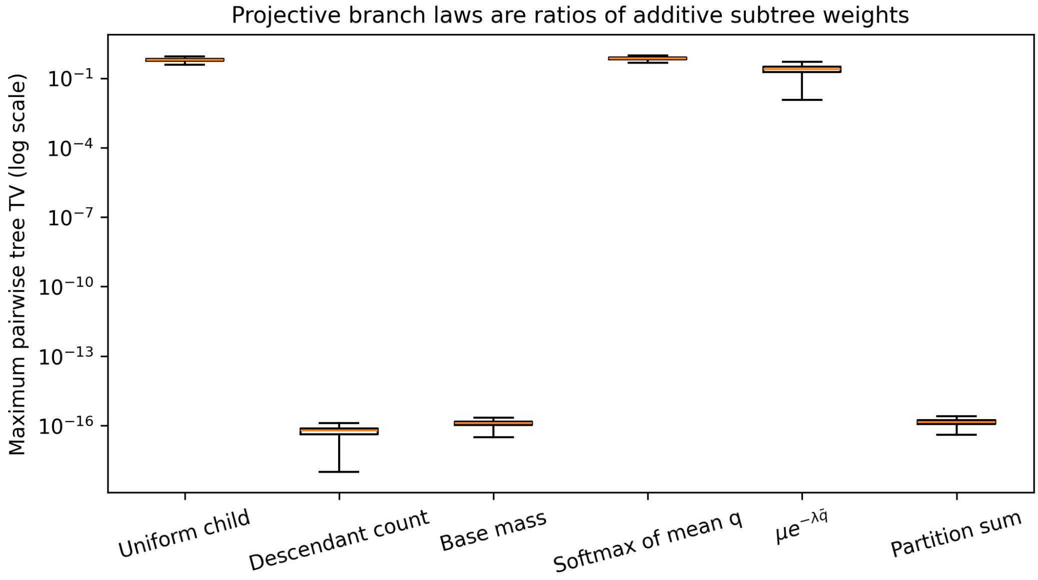

Figure 5: A33 regrouping audit. The same terminal states were embedded in different refinement trees. Only rules constructed from additive subtree weights were tree independent. Applying an exponential to a representative child mean is not additive and is therefore grouping dependent.

11 Projective branch laws

Theorem 2 (Projective ratio law)**.** *Let W be a positive finitely additive weight on a refinement hierarchy. For a parent A with disjoint children B_a,*

$$W(A) = \sum_a W(B_a). \tag{53}$$

Then

$$p(B_a \mid A) = \frac{W(B_a)}{W(A)} \tag{54}$$

is normalized and yields leaf probabilities $W(l)/W(\Omega)$ independently of grouping and refinement order. Conversely, a disjoint-regrouping-invariant path law defines such a weight up to a common scale.

A33 tested six candidate rules on 2,200 leaf configurations and six random trees per configuration. Uniform division among immediate children had median maximum pairwise tree TV 0.6181. A softmax of child mean q had median TV 0.7291, and $\mu(B)e^{-\lambda\bar{q}_B}$ had median TV 0.2471. In contrast, descendant counts, supplied base mass, and subtree partition sums were projective to floating-point precision. Figure 5 summarizes the audit.

For the exponential law, the additive weight is

$$W_\lambda(A) = \sum_{l\in A} \mu_l e^{-\lambda q_l}, \tag{55}$$

and the branch probability is

$$p(B \mid A) = \frac{W_\lambda(B)}{W_\lambda(A)}. \tag{56}$$

A global shift $q_l \mapsto q_l + c$ multiplies all subtree weights by $e^{-\lambda c}$ and cancels from (56). Maximum relative entropy with reference μ and a fixed expected-q constraint produces the same terminal law,

$$p_l = \frac{\mu_l e^{-\lambda q_l}}{\sum_k \mu_k e^{-\lambda q_k}}, \tag{57}$$

but the reference measure, constraint, and multiplier remain inputs. Projectivity selects the ratio architecture, not the terminal weights.

12 The effective score and static coarse-graining

For normalized masses $\sum_i \mu_i = 1$, define

$$Q_\lambda(q,\mu) = -\frac{1}{\lambda}\log\left(\sum_i \mu_i e^{-\lambda q_i}\right). \tag{58}$$

This is the certainty-equivalent or log-partition score associated with (57). The name *effective score* is used to avoid importing thermodynamic meaning.

For a partition into groups G, let

$$\mu_G = \sum_{i\in G}\mu_i, \qquad Q_G = -\frac{1}{\lambda}\log\left(\frac{1}{\mu_G}\sum_{i\in G}\mu_i e^{-\lambda q_i}\right). \tag{59}$$

Then

$$Q_\lambda(q,\mu) = -\frac{1}{\lambda}\log\left(\sum_G \mu_G e^{-\lambda Q_G}\right). \tag{60}$$

A34 verified (60) on 5,000 random structures with maximum error 7.11×10^{-14}. The score also obeys

$$Q_\lambda(q+c) = Q_\lambda(q) + c, \tag{61}$$

$$Q_\lambda(aq+c) = c + aQ_{a\lambda}(q), \qquad a>0. \tag{62}$$

Equation (62) is important: rescaling q changes which point on the λ-curve is needed.

Writing the cumulant generating function of q under μ gives

$$Q_\lambda = \kappa_1 - \frac{\lambda}{2}\kappa_2 + \frac{\lambda^2}{6}\kappa_3 - \frac{\lambda^3}{24}\kappa_4 + \cdots . \tag{63}$$

Two distributions with equal mean and variance can therefore have different Q_λ at finite λ. The maximum difference in the A34 matched-moment witness was 0.24259.

13 Dynamic closure obstruction

Under the deterministic centered contraction

$$q_i' = \overline{q} + a(q_i - \overline{q}), \tag{64}$$

substitution into (58) yields the exact identity

$$Q_\lambda' = (1-a)\overline{q} + aQ_{a\lambda}. \tag{65}$$

A single value Q_λ is not sufficient because the update requires $Q_{a\lambda}$. A34 constructed microdistributions with the same current mean and the same current Q_λ but different future scores. The smallest future separation in the matched construction was 0.007059, and a broader scalar-information-loss witness had median future separation 0.21753. Figure 6 displays the exact matched witness.

Exact closure for (64) is obtained by carrying the full curve

$$\lambda \longmapsto Q_\lambda, \tag{66}$$

or, equivalently, the log moment-generating function. This macrostate is generally infinite dimensional. A finite closure requires an invariant distribution family or a controlled approximation.

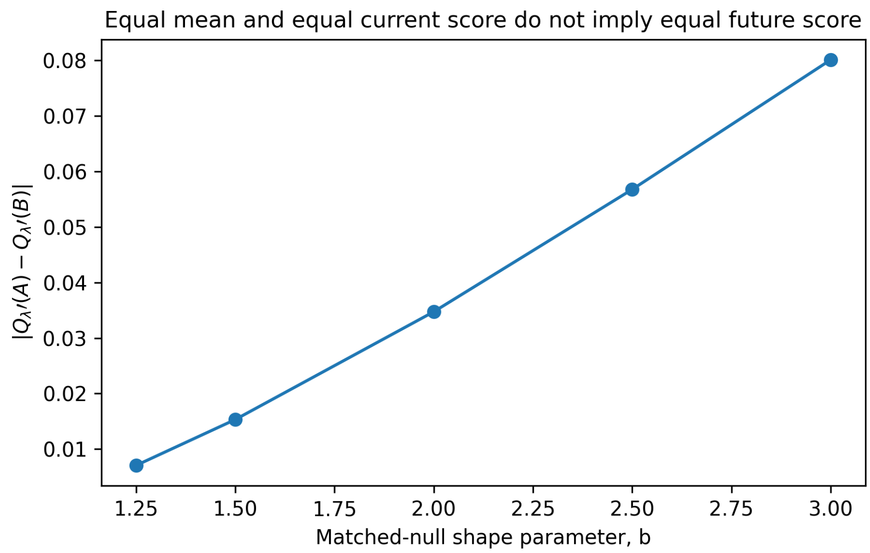

Figure 6: A34 dynamic no-go. Each pair is matched in current weighted mean and current Q_λ, yet the two future effective scores separate under the same centered contraction. Static associativity is therefore not autonomous dynamic closure.

14 Gaussian closure and the law of innovations

Let

$$X_{t+1} = b + aX_t + \xi_t. \tag{67}$$

If $X_t \sim \mathcal{N}(m_t, v_t)$ and $\xi_t \sim \mathcal{N}(0, v_\xi)$ independently, then

$$m_{t+1} = b + am_t, \qquad v_{t+1} = a^2 v_t + v_\xi, \tag{68}$$

and

$$Q_\lambda = m - \frac{\lambda}{2} v. \tag{69}$$

A35 confirmed the exact identity with maximum error 1.78×10^{-15}. This is a genuine finite-dimensional closure, but it is conditional on the Gaussian invariant family.

For independent affine innovations, cumulants satisfy

$$\kappa_r(X_{t+1}) = a^r \kappa_r(X_t) + \kappa_r(\xi_t), \qquad r \geq 2. \tag{70}$$

Gaussian innovations have no cumulants above order two, so initial non-Gaussian cumulants decay as a^{rt}. A35 compared two distributions with the same mean and variance. Their score difference decayed to 10^{-11}–10^{-13} by 30 steps, as shown in Fig. 7. The closure is asymptotic for non-Gaussian initial data, not exact at finite time.

If innovations are non-Gaussian, the stationary higher cumulants are

$$\kappa_r^* = \frac{\kappa_r(\xi)}{1 - a^r}. \tag{71}$$

Rademacher innovations adjusted to stationary variance one produced excess kurtosis −0.38007 and a maximum stationary Gaussian-closure error 0.10119. Moreover,

$$\xi \longmapsto \xi - \overline{\xi} \tag{72}$$

is a linear projection, not a Gaussianization theorem. Centered Rademacher components remained strongly non-Gaussian across dimensions 8 to 256.

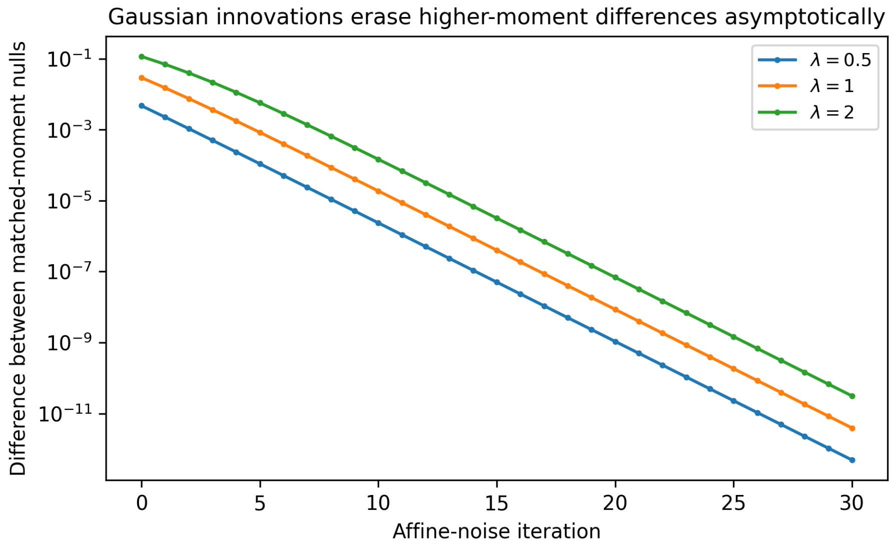

Figure 7: A35 matched-moment gaussianization. Gaussian innovations progressively erase differences generated by higher cumulants, but finite-time effective scores remain distinct. Larger λ retains greater tail sensitivity.

14.1 What can select Gaussian noise?

Several principles were audited separately. With fixed mean and variance, the Gaussian uniquely maximizes differential entropy [14, 24]. Exact iid stability

$$\frac{X_1 + X_2}{\sqrt{2}} \overset{d}{=} X \tag{73}$$

with finite variance also selects the Gaussian. Rotational invariance plus independent components gives a Maxwell-type characterization. However, centering, rotational invariance alone, projectivity, and infinite divisibility are non-selective. Cauchy laws are non-Gaussian stable when finite variance is dropped, and Gaussian scale mixtures provide rotationally invariant projective non-Gaussian families [25].

A36's original finite-size KS gate failed for Rademacher lattice sums at block size 128. A36.1 retained the failure record and confirmed the conditional CLT by testing

$$\phi_n(t) = \phi(t/\sqrt{n})^n \longrightarrow e^{-t^2/2} \tag{74}$$

uniformly on a frozen compact interval. The corrected result supports a conditional theorem; it does not derive independence or finite-variance innovations for the Modal Field.

15 Noise-law universality and its limits

For the stationary affine process

$$X_{t+1} = aX_t + \xi_t, \qquad |a| < 1, \tag{75}$$

the characteristic function is

$$\phi_X(t) = \prod_{k=0}^{\infty} \phi_\xi(a^k t). \tag{76}$$

Matching innovation means and variances fixes the stationary mean and variance, but (71) shows that higher cumulants remain noise-law dependent. A37.1 compared Gaussian, Rademacher, uniform, Laplace, and Student-t_8 innovations with identical stationary variance. The stationary excess-kurtosis spread was 0.95017.

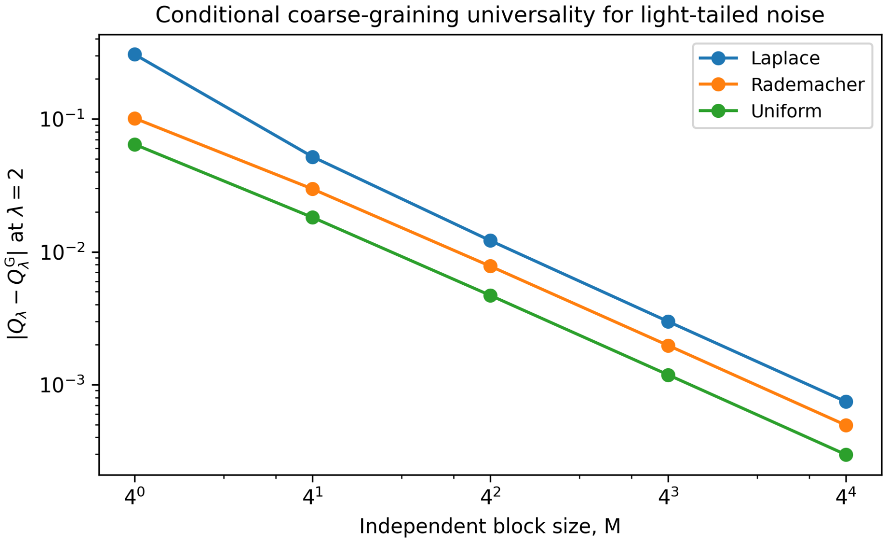

Figure 8: A37.1 conditional universality. Independent block sums of light-tailed stationary laws converge toward the Gaussian effective score. Student-t is absent because its exponential score is undefined at every finite block size.

The effective-score spread among light-tailed laws increased with λ:

$$\Delta Q_{0.5} = 0.00498, \quad \Delta Q_1 = 0.04083, \quad \Delta Q_2 = 0.40743. \tag{77}$$

Student-t_8 has finite variance but no finite nonzero moment-generating function, so Q_λ is not finite. This is a sharp warning: finite variance is not enough for a partition-sum observable.

For M independent stationary copies,

$$Y_M = \frac{1}{\sqrt{M}} \sum_{i=1}^{M} X_i, \tag{78}$$

and

$$\log \mathbb{E}[e^{sY_M}] = M \log M_X(s/\sqrt{M}). \tag{79}$$

With controlled exponential moments, the block effective score converges to the Gaussian value. At $M = 256$, the maximum light-tailed error across the audited laws and λ values was 7.43×10^{-4}. Figure 8 shows the convergence at $\lambda = 2$.

Weak convergence alone does not guarantee convergence of exponential moments [26, 27]. Dependence can also block Gaussian universality. A common Rademacher shock with correlation fraction $\rho = 0.25$ survived variance-normalized aggregation; at block size 1024 the excess kurtosis was -1.98833, close to the Rademacher limit -2. The law of noise can be omitted from the fundamental description only after a relevant universality class has been derived for the actual coarse-graining operation.

16 Observable-relative coarse-graining

16.1 Translation-covariant decomposable means

Consider a weighted quasi-arithmetic mean

$$M_f(q; \mu) = f^{-1}\left(\frac{\sum_i \mu_i f(q_i)}{\sum_i \mu_i}\right). \tag{80}$$

Within the audited regular class, decomposability plus translation covariance

$$M_f(q + c; \mu) = M_f(q; \mu) + c \tag{81}$$

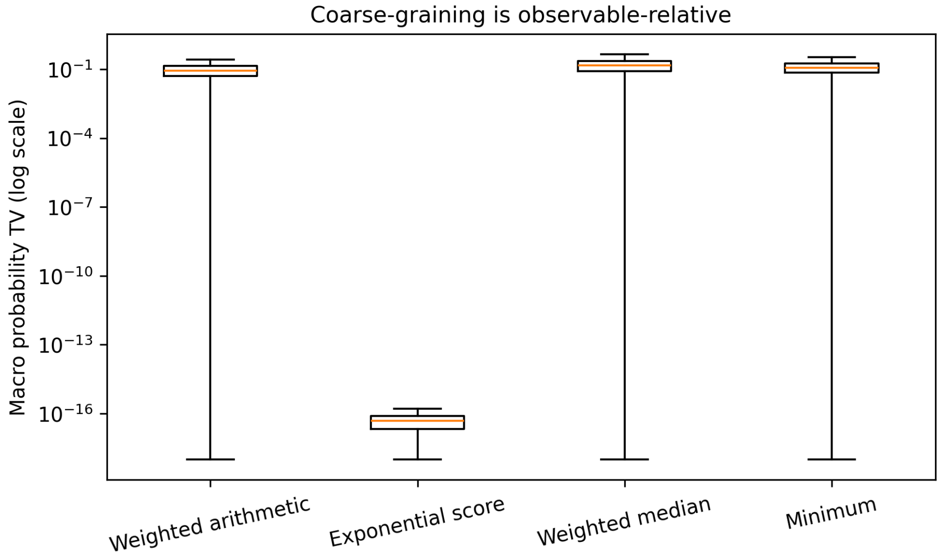

Figure 9: A38.1 observable-relative coarse-graining. The exponential effective score is an exact static message for an exponential microscopic kernel. Other summaries can be internally valid yet fail to preserve this observable.

restricts the generator, up to equivalent affine transformations, to the linear and exponential families. This is consistent with the broader theory of quasi-arithmetic means [28, 29].

The weighted arithmetic mean

$$\overline{q}_\mu = \frac{\sum_i \mu_i q_i}{\sum_i \mu_i} \tag{82}$$

is the correct sufficient statistic for the first moment. The exponential score (58) is the correct sufficient statistic for the exponential weight (55). The minimum is associative and translation covariant but extremal and mass insensitive. The weighted median is refinement consistent but is not closed by the pair "block mass + block median." A38.1 supplied the exact counterexample

$$\mathrm{med}\{(0, 0.4), (2, 0.1), (1, 0.5)\} = 1, \quad \mathrm{med}\{(0, 0.5), (1, 0.5)\} = 0, \tag{83}$$

where the second median is computed from block summaries.

Figure 9 shows that only the exponential score preserved the audited exponential kernel. The weighted arithmetic mean had median macro-probability TV 0.08755; the exponential score's maximum TV was 2.19×10^{-16}.

The correct conclusion is not that Q_λ is universally preferred. It is that coarse-graining is defined relative to a declared observable. The first moment, an exponential partition weight, and an extremal statistic require different sufficient messages.

16.2 Graph dynamics and occupancy obstruction

Let microscopic states be partitioned into regions A, B. The transition from source region A to target region B is

$$K_{AB}^{\mathrm{macro}} = \sum_{i \in A} \rho(i \mid A) \sum_{j \in B} K_{ij}, \tag{84}$$

where $\rho(i \mid A)$ is the conditional source occupancy. The same microscopic K and the same partition can therefore induce different macro kernels when the internal occupancy changes. A38.1 found median mean row-wise TV 0.10122 between two occupancies on the same graph, as shown in Fig. 10.

If a global occupancy π is supplied, the joint flow (11) aggregates exactly:

$$F_{AB} = \sum_{i \in A} \sum_{j \in B} F_{ij}, \qquad \pi_A = \sum_{i \in A} \pi_i, \tag{85}$$

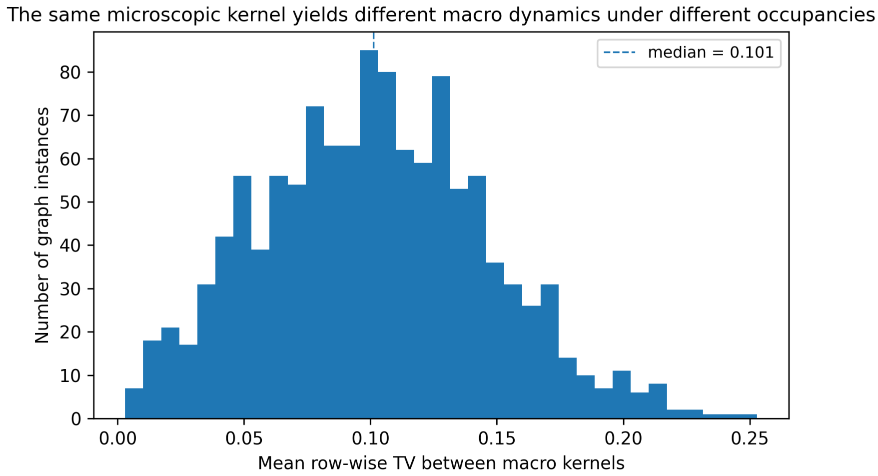

Figure 10: A38.1 source-occupancy obstruction. The histogram reports the mean row-wise total-variation distance between macro kernels obtained from the same microscopic kernel and partition but different source occupancies.

followed by

$$K^{\text{macro}}_{AB} = \frac{F_{AB}}{\pi_A} \quad \text{for } \pi_A > 0. \tag{86}$$

The maximum flow-aggregation error was 8.88×10^{-16}. This is closely related to the dependence of Markov aggregation on initial or stationary distributions and lumpability conditions [18, 20, 21].

17 The current Modal Field architecture

The audits produce the dependency structure

$$R \longrightarrow \preceq_R, \tag{87}$$

meaning that a generic directed relation canonically yields an internal order after SCC condensation. However,

$$R \not\Longrightarrow \text{unique clock, calibrated duration, or measure.} \tag{88}$$

An independent quantitative mark can supply relative distinctions,

$$(R, q) \longrightarrow \text{relative quantitative observables}, \tag{89}$$

but the gauge (3) blocks an absolute offset and, through (6), an absolute global length scale. Refinement consistency introduces additive multiplicity,

$$(R, q, \mu) \longrightarrow \text{projective static weights}, \tag{90}$$

while a declared response law introduces transitions,

$$(R, q, \mu, \lambda) \longrightarrow K. \tag{91}$$

Finally, unique macroscopic transport requires occupancy or flow,

$$(K, \pi) \longrightarrow F = \operatorname{diag}(\pi) K. \tag{92}$$

The present minimal candidate is therefore

$$\boxed{\mathcal{M}_{\min} = (R, q, \mu, K, \pi; \mathfrak{F})}, \tag{93}$$

with the explicit warning that minimality has not been proven. The programme's next task is compression: determine whether μ, K, and π can arise from a smaller common principle rather than being appended independently.

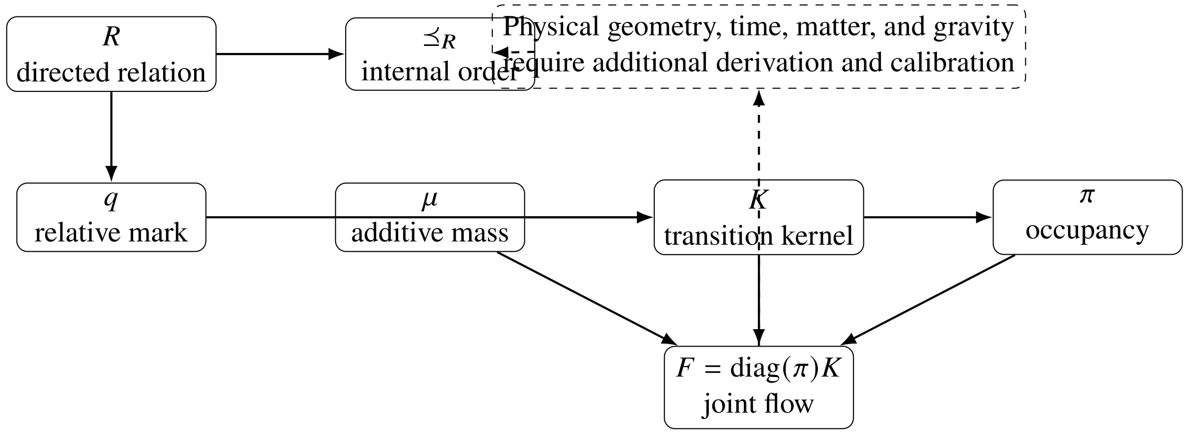

Figure 11: Current architecture of the Modal Field research programme. Solid arrows denote constructions established under declared assumptions. Dashed arrows denote the open emergence problem; no physical spacetime identification is asserted.

18 What has been established

The strongest positive results are the following.

P1. Generic finite directed relations admit a canonical condensation partial order.

P2. The q field can carry genuine information beyond order if its conditional law is independently marked.

P3. Relative q contrasts are operationally identifiable from a declared local transition law with calibrated coupling.

P4. Additive mass gives exact consistency under descriptive clone refinements.

P5. Consistent child fractions define probability measures on infinite refinement path spaces; terminal leaves are unnecessary.

P6. Regrouping-invariant branch laws are ratios of additive subtree weights.

P7. The partition-sum weight $W_\lambda = \sum \mu e^{-\lambda q}$ is exactly projective and gauge invariant at the probability level.

P8. The effective score Q_λ is an associative static coarse variable when paired with additive mass.

P9. Gaussian affine dynamics has an exact closure in mean and variance, conditional on Gaussianity.

P10. Joint flows aggregate exactly when occupancy is supplied.

The strongest no-go results are:

N1. Order alone does not select a unique scalar clock or calibrated duration.

N2. Global interval signatures are not sufficient for exact manifoldlikeness.

N3. Order and cardinality cannot identify monotone marginal or conformal-volume representatives.

N4. Endogenous order-measurable dynamics cannot break that identifiability symmetry.

N5. Gauge covariance leaves the absolute q offset and global D_{eff} scale undetermined.

N6. Locality, relabeling symmetry, and scale freedom do not select a unique q-dependent kernel.

N7. The exponential functional form does not determine λ.

N8. Bare (R, q) data cannot decide whether exact clones are descriptive or ontically distinct.

N9. Projectivity does not select terminal weights or split fractions.

N10. A fixed scalar Q_λ is not generally dynamically closed.

N11. Centering does not select Gaussian noise.

N12. Mean and variance matching do not make stationary or operational observables noise-law universal.

N13. A microscopic kernel and partition do not determine a unique macro kernel without source occupancy.

19 What the Modal Field is and is not

19.1 It is

The Modal Field is a proposed mathematical language for pre-categorical structure. It is useful for proving necessity statements: if a theory begins only with relations, then certain quantitative structures cannot be recovered without extra assumptions. It also supplies projective constructions that remain coherent under finite or infinite refinement. In this sense, the framework is already a substantive research programme: it produces exact theorems, counterexamples, reproducible audits, and a constrained architecture.

19.2 It is not

It is not presently a quantum theory, a theory of gravity, or a derivation of Einstein's equations. D_{eff} is not a measured spatial distance. Q_λ is not a temperature-dependent free energy. μ is not yet a physical volume element. π is not yet identified with matter density, Born probability, or realized existence. The framework has not derived a Lorentzian metric, curvature, matter coupling, quantum amplitude, calibrated clock, or experimental prediction.

These statements are not qualifications added to weaken the proposal. They define the actual scientific target. A pregeometric theory should explain how such identifications arise rather than placing them into the notation at the outset.

20 Next programme: from architecture to compression

The immediate continuation is A39, an audit of the origin and dynamics of occupancy. Candidate identifications include

$$\pi = \frac{\mu}{\sum_i \mu_i}, \tag{94}$$

$$\pi_{t+1} = \pi_t K_t, \tag{95}$$

$$\pi^* = \pi^* K, \tag{96}$$

but each raises distinct questions. Equation (94) identifies possibility mass with realized occupancy and may be semantically unjustified. Equation (95) requires an initial condition and a time-index interpretation. Equation (96) may be non-unique or fail to exist for reducible kernels. If K depends on π, a nonlinear fixed-point problem appears, with possible multiple solutions and symmetry breaking.

A broader compression programme should test whether a single conserved or variational object can generate μ, K, and π. One candidate direction is an action or divergence functional on flows:

$$\mathcal{A}[F; q, \mu] = \sum_{ij} F_{ij} \log \frac{F_{ij}}{\pi_i \mu_{ij}} + \lambda \sum_{ij} F_{ij} q_{ij}, \tag{97}$$

subject to conservation constraints. Equation (97) is presented only as a future test template. It must not be treated as a derived law. The research question is whether such a principle can reproduce the accepted projective structures without importing the desired answer through its reference measure or constraints.

Additional priorities are:

(i) derive or falsify a native origin for q;

(ii) determine whether λ runs under a physically justified coarse-graining;

(iii) establish a non-circular relation between multiplicity μ and occupancy π;

(iv) identify local observables whose statistics can be compared with external data;

(v) revisit geometric eligibility only after dynamic closure and scale calibration are secured.

21 Discussion

The main scientific value of the Modal Field is not that it has already explained spacetime. Its value is that it converts a broad metaphysical intuition into a sequence of answerable mathematical questions. The framework has survived by becoming more constrained. Relations were sufficient for internal order but insufficient for measure. A quantitative mark supplied relative information but left a gauge. Additive mass repaired refinement but introduced multiplicity semantics. Projective branch laws fixed an architecture but not terminal weights. The effective score solved static aggregation but failed dynamic closure. Gaussian closure worked exactly but only under a non-derived innovation law. Coarse-grained dynamics finally required occupancy.

This progression can be read pessimistically as a growing list of primitives. A more constructive reading is that the audits have exposed the exact interfaces at which a compact principle would need to act. The current architecture is therefore a target for reduction. A successful fundamental theory should explain why the objects in (93) are not independent accidents.

The programme also clarifies how claims of emergence should be assessed. A global statistical resemblance is not an embedding theorem. A gauge-invariant shape is not an absolute metric. A projective measure is not a physical volume unless its operational meaning is specified. A closed form is not a dynamical closure unless future values are functions of the retained macrostate. These distinctions are broadly relevant to pregeometry and emergent-spacetime research.

22 Conclusion

The Modal Field provides a technically coherent proposal for studying pre-categorical relational structure. Its present content can be summarized by three statements.

First, relations generate less than is often assumed. They can generate a canonical internal order, but not a unique clock, quantitative measure, calibrated scale, or autonomous macrodynamics.

Second, the missing structures are not arbitrary decorations. The audits identify their mathematical roles. The q field supplies relative quantitative information; μ supplies additive multiplicity and projective refinement; K supplies a declared response law; π supplies the occupancy needed for macroscopic transport. Exact theorems show how these objects compose and where they fail to be sufficient.

Third, the framework remains an open theory proposal rather than a final physical theory. Its strongest current claim is not that spacetime has been derived, but that a rigorous route toward such a derivation has been narrowed. Any future completion must reduce the primitive count, derive the coupling and noise structure, establish dynamic closure, calibrate scale and time, and connect modal observables to experiment. The programme is worth pursuing because it now makes enough precise statements to be wrong, corrected, and improved.

A Audit chronology and verdicts

Table 2: Condensed chronology of the computational and theorem-driven audits used in this synthesis (Part I). "Pass" means the frozen target of the specific audit was met; it never means that the full physical theory was established.

Audit	Question	Main result	Status
A3	Degree sufficiency	Degree fibers retain extensive relational variation.	Exact
A4	Entropy-budget selection	Tested graph controls are not uniquely selected within matched fibers.	Negative
A5	Finite projectivity	Induced-subgraph projectivity leaves the full top-level simplex.	Exact
A6	Symmetry selection	Reversal/complement symmetries reduce but do not select a unique law.	Exact
A8	Internal order	SCC condensation yields 5,234 labeled poset codes from all five-node digraphs.	Exact
A9	Scalar clocks	Depth, balance, and linear-extension clocks are monotone but non-unique.	Exact
A10	Durations	Longest-chain duration is strongest ordinal candidate but uncalibrated and non-coarse-invariant.	Exact
A11	Broad geometry discrimination	Frozen feature set failed two of six broad gates.	Fail
A12	Targeted 2D discrimination	Minkowski orders separated from ordering-fraction-matched percolation nulls.	8/8 pass
A13	Analytic interval profile	Analytic-reference test failed two of nine gates.	7/9
A14	Covariance-aware profile	Minkowski holdouts accepted and declared nulls rejected.	9/9 pass
A15	Sufficiency attack	Certified local obstructions passed the A14 global statistic.	Counterexample

Table 3: Condensed chronology of the audits (Part II).

Audit	Question	Main result	Status
A16	Local/global combination	One clustered exact-product-order family defeated the frozen complementarity gate.	9/10
A17	Multiscale similarity	Detects some heterogeneous regimes but not compact density clustering.	6/9
A18.2	Single-order identifiability	Monotone marginal transforms preserve the order exactly.	9/9 pass
A19	Ensemble identifiability	Fixed copula implies identical finite-order laws for all continuous marginals.	10/10 pass
A20	Minimal symmetry breaking	Independent or ordinal marks fail; calibrated quantitative marks can break symmetry.	10/10 pass
A21	Endogenous dynamics	Order-measurable dynamics preserves the latent equivalence class.	11/11 pass
A22	Primitive admissibility	A calibrated binary mark is sufficient mathematically, but its scale is external.	13/13 pass
A23	RZS q audit	Relative q can add information; global shift and rate-time degeneracies remain.	13/13 pass
A24.1	Operational q	Local log-odds recover q contrasts conditional on calibrated β.	12/12 pass
A25	Non-circular couplings	Scale-free local kernels exist, but infinitely many laws satisfy the same symmetries.	11/11 pass
A26	Kernel principles	Difference-only odds selects an exponential family; strength remains free.	12/12 pass
A27	Status of λ	Identifiable after fixed standardization, estimable from data, not derived.	11/11 pass

Table 4: Condensed chronology of the audits (Part III).

Audit	Question	Main result	Status
A28	Score no-go	Extension stability, affine invariance, continuity, and cardinality force a trivial score.	10/10 pass
A29.1	Contextual refinement	Additive mass restores exact clone consistency; zero-variance boundary remains singular.	11/12 fail
A30	Origin of mass	Bare (R, q) cannot distinguish descriptive clones from ontic multiplicity.	11/11 pass
A31	Finite refinement	Leaf-weight extension is unique and path independent, conditional on leaf ontology.	10/10 pass
A32	Infinite refinement	Projective measure exists without terminal leaves; atomicity is independent of terminality.	10/10 pass
A33	Branch fractions	Regrouping invariance forces ratios of additive subtree weights.	12/12 pass
A34	Effective score	Static aggregation is exact; a fixed scalar score is not dynamically closed.	10/10 pass
A35	Gaussian closure	Mean–variance closure is exact only under Gaussian state and innovations.	12/12 pass
A36/A36.1	Noise selection	Gaussian law requires strong additional axioms; corrected CLT diagnostic passes conditionally.	Corrected
A37/A37.1	Noise universality	Partial universality requires independence/mixing and exponential-tail control.	Corrected
A38/A38.1	Coarse-graining	Sufficient statistics are observable-relative; macro dynamics requires occupancy.	Corrected

B Selected exact numerical results

Table 5: Representative numerical results quoted in the main text. Values are reproduced from the generated JSON/CSV audit artifacts.

Quantity	Value
All labeled five-node loopless digraphs enumerated	1,048,576
Distinct labeled condensation posets	5,234
Single-edge flips changing labeled condensation	0.29296875
Comparable triples with longest-chain additivity	0.9892818864
A24.1 mean contrast-recovery RMSE	0.0159935
A27 median TV between distinct standardized λ kernels	0.2382046
A31 maximum tree path-independence error	4.55×10^{-13}
A32 atomic nonterminal path mass at depth 160	0.5031056
A33 uniform-child median tree TV	0.6180556
A34 median scalar-loss future difference	0.2175321
A35 stationary Rademacher excess kurtosis	−0.3800669
A37.1 stationary $Q_{\lambda=2}$ spread	0.4074312
A38.1 median occupancy-induced macro row TV	0.1012237

C Reproducibility statement

The manuscript was generated from the exact local audit artifacts A3–A38.1. Each computational stage uses explicit random seeds and writes machine-readable CSV and JSON outputs. The figures in this article were generated directly from the following files:

- A14 sample scores and A15 certified counterexample scores;
- A24.1 contrast recovery;
- A29.1 refinement audit;
- A32 atomicity without terminality;
- A33 tree regrouping audit;

- A34 matched-score dynamic witness;
- A35 matched-moment gaussianization;
- A37.1 independent-block universality;
- A38.1 exponential-kernel preservation and source-occupancy obstruction.

No public archive identifier or external replication is claimed in this version. A submission version should deposit the source scripts, frozen environments, data tables, and figure-generation code in a permanent repository before peer review.

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